



BY AMETECs

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Development Session Report — 11–12 August 2026

Progressive Lifetime Pricing · Live-Server Triage · Client-wise SMTP & Forgot-Password · Admin Cleanup Tools

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1. Executive Summary

This session delivered **nine committed work packages across both SmartEPT repositories** — SmartEPT Central (`smartept-central` , the commercial platform at `smartept.com`) and the SmartEPT Product console (`smartept` , the client console at `admin.smartept.com`) — plus two infrastructure repairs on the development PC and a guided triage of the live VPS health check.

The headline items: the **On-Premise Lifetime licence pricing model was changed from flat band pricing to progressive milestone interpolation** (backend, all checkout surfaces and the public landing calculator, verified 64/64 against the owner's acceptance matrix); a **root-cause login bug that had silently made every seeded product-console login unusable was found and fixed** (double password hashing); the product gained **client-wise SMTP configuration with global fallback** and a **forgot-password email-OTP flow**; Central gained **super-admin Delete controls** for test-data cleanup and a **live-coupon-aware buy page**; and the **hourly dunning crash on the live server was root-caused and fixed** (third ENUM strike — column converted to VARCHAR).

Additionally, **smartept.com was verified by Razorpay** during this session — the payment remediation work is complete and live keys are configured in Central Settings.

Commit ledger (chronological):

#	Repo	Commit	What
1	Central	<code>e6f5839</code>	Progressive/proportional On-Premise Lifetime pricing (milestone interpolation)
2	Central	<code>d8279a5</code>	Landing pricing slider — perpetual price now progressive
3	Central	<code>a03f228</code>	Remove ™ symbol from SmartEPT branding (© kept)
4	Central	<code>73888fc</code>	Super-only Delete for quotes/unpaid orders + invoices
5	Central	<code>72c0059</code>	Coupon box on /buy hidden when no coupon is live
6	Central	<code>2d9d8a6</code>	Licence signing keypair committed (owner override — repo must stay private)
7	Central	<code>385836a</code>	Dunning crash fix — <code>tenants.status</code> ENUM → VARCHAR(20)
8	Product	<code>22ed12e</code>	Seeder double-hash fix + client-wise SMTP + forgot-password OTP
9	Central	<code>efd4797</code>	(Owner's own commit — housekeeping of stale lock files)

2. Complete Change List at a Glance

Fixed (bugs):

1. Product console seeded logins never worked — password double-hashed in the seeder (local AND live affected).
2. Hourly `smartept:dunning` job crashed on live every hour — `tenants.status` ENUM lacked the value `purged` (strict-MySQL 1265).
3. Development PC's new PHP 8.5.9 had **no php.ini at all** — no extensions loaded, `migrate.bat` failed with "could not find driver".
4. Landing page pricing slider showed flat band prices after the pricing model changed (display inconsistency with /buy).

Changed (behaviour):

5. Lifetime licence pricing: flat band price → progressive milestone interpolation (the session's core work).
6. AMC now charged at 18% of the **interpolated** prevailing price (owner decision), not the band-max price.
7. Open-ended Custom Quote pricing band now stores price **NULL** (never 0) — a ₹0 lifetime licence is impossible by construction.
8. Coupon input on /buy appears only when at least one coupon is genuinely usable (enabled + in validity window + uses left).
9. TM symbol removed from all SmartEPT branding (nothing trademarked yet); © 2026 notices kept.
10. Licence signing keypair now travels via git (explicit owner override of the never-in-git rule; repo must remain private forever).

New (features):

11. `PricingService::calculateLifetimeLicencePrice()` — the single reusable progressive-pricing engine.
12. Super-admin **Delete** on quotes/unpaid orders and on invoices (rule-guarded, confirmed, audit-logged).
13. **Client-wise SMTP** in the product console: each client company configures its own relay; global default fallback; Send-test buttons.
14. **Forgot-password** on the product console login (email OTP, all roles including employees).
15. Full acceptance-test suite `PerpetualProgressivePricingTest` (Central) covering the owner's pricing matrix.

Improved (hardening/validation):

16. Pricing-band editor now rejects overlaps, gaps, duplicates, min>max, non-positive or decreasing prices, and misplaced open-ended bands.
17. `BillingService::createOrder()` hard-throws on any custom/below-minimum perpetual order — the last door against a free licence.

18. Below-minimum user counts get a clear validation message sourced from configuration ("Minimum On-Premise licence capacity is N users.").
19. All "more than 5,000 users" hard-coded messages replaced with dynamic values derived from the configured bands.

3. Progressive Lifetime Pricing — Full Detail

What. Each configured band's price now means "one-time price AT that band's maximum-users milestone", not a flat price for every count inside the band. Counts between two milestones are priced by straight-line interpolation:

$$\text{Price} = \text{PrevMilestonePrice} + (\text{Users} - \text{PrevMilestoneUsers}) \times (\text{CurrMilestonePrice} - \text{PrevMilestonePrice}) \div (\text{CurrMilestoneUsers} - \text{PrevMilestoneUsers})$$

The first band is the **minimum licence package** — flat price, never reduced. Below the first band's minimum → validation error. Above the last priced milestone → **Custom Quote**, never an automatic price.

Why. Under flat banding, 151 users jumped straight to the 200-user price (₹90,000). The customer should pay proportionally for additional users: with milestones 150 = ₹75,000 and 200 = ₹90,000 the increment is ₹15,000 ÷ 50 = ₹300/user, so 151 users = ₹75,300, 175 = ₹82,500, 200 = ₹90,000 exactly.

How.

- **Engine** (`app/Services/PricingService.php`): new `calculateLifetimeLicencePrice(Plan, users)` returning requested users, price, previous/next milestone, per-user rate, `custom` and `below_min` flags, configured minimum and last priced milestone. Integer arithmetic with **one final round** — configured milestones always land exactly on their configured price; zero cumulative float drift. `perpetualQuote()` and `amcQuote()` rewired onto it. Fully dynamic: the Super Admin re-prices milestones and every rate recalculates — nothing hard-coded.
- **Guards**: `BillingService::createOrder()` throws on custom/below-min/empty perpetual quotes; every controller (public /buy ×2, client portal quote + order, admin quote + order + prospect quote) distinguishes below-minimum (422 with configured minimum) from custom quote (message derived from the last priced milestone).
- **Custom band storage**: migration `2026_08_11_000100` relaxes `plan_perpetual_bands.price_inr` to nullable and converts any legacy 0-priced open band to NULL. Non-destructive; no pricing rows touched.
- **Band-editor validation** (`ConfigApiController::savePerpetualBands`): overlaps, gaps, duplicates, min>max, missing/zero/negative price on a priced band, decreasing milestone prices, and open-ended band anywhere but last are all rejected with specific messages. The open band saves with price NULL.

- **Super Admin UI:** price column re-headed "₹ price at Max users (milestone)", explanatory note added ("Prices between user milestones are calculated proportionately. The first band is the minimum licence price."), blank-Max rows show **Custom Quote** with the price input disabled.
- **Frontend calculators** (display only — the backend always reprices on submit): /buy page, trial/sign-up page and client portal all mirror the interpolation, fed live from the admin-saved bands via the public plans API (5-minute cache). The portal's hard-coded 5,000-user check was removed — the server decides.
- **Landing page** (`public/landing.html` , commit `d8279a5`): the npx pricing widget gained `npxPerpPrice()` with the same maths. Verified identical to backend: $41 \rightarrow ₹33,250 \cdot 47 \rightarrow ₹37,750 \cdot 61 \rightarrow ₹44,400$ (matches the /buy screenshots taken during testing). **Owner action:** run `php artisan landing:import` once so the Landing CMS keeps this on a future publish.

Verification. The complete owner acceptance matrix was executed against the real service code in a sandbox harness — **64/64 passed:** 5/10/15 $\rightarrow$ ₹15,000 flat; 16 $\rightarrow$ ₹15,667 (single final round of ₹666.67/user); 30/50/100/150/200 exactly on milestone; 151 $\rightarrow$ ₹75,300; 160 $\rightarrow$ ₹78,000; 175 $\rightarrow$ ₹82,500; 199 $\rightarrow$ ₹89,700; 201/500/99,999 $\rightarrow$ Custom (price NULL); 1/4 $\rightarrow$ below-minimum; re-price test (150 = ₹80k, 200 = ₹1L $\Rightarrow$ 151 = ₹80,400 at ₹400/user) proving full dynamism; per-user drift check across an entire fractional segment; ₹0-impossibility checks. The same matrix is codified permanently in `tests/Feature/PerpetualProgressivePricingTest.php`.

Commercial note. Prices INSIDE former bands are now lower than the old flat model by design (e.g. 40 users: was ₹50,000 flat under old seeded bands, now interpolated). Milestone prices are re-tunable at any time in Super Admin $\rightarrow$ Plans & Pricing.

4. Live VPS Health Triage — Four Issues, One by One

The live System Health check (`smartept.com/admin` $\rightarrow$ Help & Troubleshooting) showed one red and three amber items. Resolved with the owner interactively:

4.1 Licence signing key missing on live (RED — blocked all .lic issuing). The RSA-2048 private key existed only on the dev PC because EPT-29 (`3c2b497`) deliberately gitignored it per the Distribution & Licensing Standard. The owner explicitly overrode the rule: commit `2d9d8a6` places `license_private.pem` + `license_public.pem` in the repo so the VPS receives them via `git pull`. **Standing consequence: the SmartEPT_central repository must remain PRIVATE forever** — anyone with repo access can sign licences. On the VPS after pull: `chmod 600` the private key, `chown www-data:www-data` both.

4.2 Hosted-console link not configured (amber — blocked cloud provisioning only). Cloud console = `admin.smartept.com` on the same VPS. Steps issued: generate two secrets (`openssl rand -hex 32` twice); in live Central Settings set Product URL `https://admin.smartept.com` , Provisioning secret = A, SSO secret = B (screenshot confirmed the owner filled these); in the

product's live `.env` set `PROVISION_SECRET=A`, `SSO_SHARED_SECRET=B`, then `config:clear` + reload php-fpm. Secrets must match pairwise.

4.3 WhatsApp (Interakt) not configured (amber — email already covers all notifications).

The owner's Interakt API key is ready; instructions: Settings → WhatsApp API → paste key (base64, exact), sender number digits-with-country-code, WABA ID optional, API URL blank (defaults to the official Interakt endpoint), status **active**, Save, re-run checks. Note: since the key passed through chat, regenerate it in Interakt if there is ever any doubt.

4.4 Two errors per hour (amber) — root-caused and FIXED (385836a).

`RunDunning::enforcePurgeWindows()` sets `tenants.status = 'purged'` for expired tenants past their purge window, but the column's ENUM never included `purged` → strict-MySQL error 1265 killed the whole hourly dunning run at that step (tenant #7, every hour on the hour; the second error is the scheduler reporting the failure). This is the **third** ENUM to break a live path. Fix: migration `2026_08_11_000300` converts `tenants.status` to `VARCHAR(20)` (sqlite-guarded so the test suite is unaffected), per the standing prefer-VARCHAR lesson. After deploy, tenant #7 purges cleanly on the next hourly run.

5. Product Console — Login Bug, Client-wise SMTP, Forgot-Password (22ed12e)

5.1 Root-cause login bug (fixed). `super@smartept.io` could not sign in with the seeded password `password` — on local AND live. Cause: the `User` model declares `password => 'hashed'` (Laravel auto-hashes on assignment) while `DatabaseSeeder` ALSO called `Hash::make('password')` — the hash got hashed a second time, so the stored value matched nothing. Every seeded account (`super@smartept.io`, `admin@ameteecs.io`, `manager@`, `hr@`, `compliance@`) was unloginable since day one. Fix: the seeder now passes the plain value (the cast hashes exactly once). Existing installs unlock with one tinker line (given to the owner; run on local and VPS): set a strong password immediately — a live super-admin with password `password` would be the platform's biggest hole. **Team lesson: with the 'hashed' cast, never pre-hash; and `Model::where()->update()` bypasses casts — use `->first()->forceFill([...])->save()` for passwords.**

5.2 Client-wise SMTP (new). Per the owner's decisions: OTP-based resets; COMPANY_ADMIN + Super can edit; fallback to the global relay.

- `companies.mail_settings` (JSON, new migration `2026_08_12_000100`): host, port, username, **encrypted** password, encryption (TLS/SSL/none), from-address, from-name. Clearing the host removes the override entirely.
- Global default relay stored in the product's Settings table (`mail_host...mail_from_name`, password encrypted) — the install's default, Super Admin only.

- `MailService` now resolves **every** outbound mail: company's own SMTP → global Settings SMTP → `.env` mailer, with the right From identity. Alerts, credential mails and OTPs all pass through this one door; every attempt still lands in `mail_logs`.
- Console UI: Ops screen gained an **Email / SMTP** card — "My company's email (SMTP)" section for company admins and "Global default relay" section for Super, each with **Send test email** (the test reports the relay's actual error message on failure) and blank-password-keeps-saved semantics. All saves audit-logged.
- Honest scope note: a "pick any company and edit its SMTP" picker for Super was NOT built — each company admin manages their own; Super manages the global. Can be added on request.

5.3 Forgot-password OTP (new). "Forgot password?" link on the console login card (works on /admin and on branded /slug logins, which stay tenant-locked):

- `POST /api/auth/forgot/request-otp` (throttle 5/min): always answers neutrally (no account enumeration); for a real, non-disabled account it emails a 6-digit code — via that user's company SMTP chain — valid 10 minutes.
- `POST /api/auth/forgot/reset` (throttle 10/min): verifies the hashed OTP (max 5 attempts), enforces min-8 password, sets it through the cast (single hash), clears `must_change_password`, deletes the OTP. Both steps audit-logged (`PASSWORD_RESET_REQUESTED` / `_COMPLETED`).
- Works for every role **including employees** (employee-portal users live in the same users table; they need an email on file).

6. Central Admin — Cleanup Tools & Coupon Behaviour

6.1 Delete controls (73888fc). The Orders & Payments and Invoices screens had no way to remove test garbage. Now (super admin only, red links, confirm dialogs, audit-logged):

- **Quotes / unpaid orders:** Delete appears only when the payments ledger shows ₹0 received AND no licence is linked; deleting removes the order, its unpaid invoice and (empty) payment rows in one transaction. **An order with even one rupee on the ledger can never be deleted** — the server refuses (refund/credit-note path instead). The rev186 ledger principle is fully preserved.
- **Invoices:** any invoice deletable by Super with a strong warning — GST invoice numbers are FY-consecutive and never reused, so a mid-series delete leaves a permanent gap. This is a **pre-launch decision** for test-data cleanup; post-launch the credit-note path is the compliant way (revisit then — e.g. restrict to newest-of-FY or disable).
- No Edit was added (owner choice): wrong quote = delete + re-create via the existing one-screen flow.

6.2 Coupon box visibility (72c0059). The /buy page showed the coupon input even when every coupon was disabled. New `Coupon::anyLive()` (enabled + within valid_from/valid_until + uses remaining) drives server-side rendering of the box; JS wiring is null-guarded and the exclusive-offer auto-check skips when the box is absent. Redemption checks unchanged. (The client portal's coupon field was left as-is per scope; same one-line treatment available on request.)

6.3 Branding (a03f228). Both live instances of "SmartEPT™" (buy-page footer, client sign-in footer) are now plain "SmartEPT"; © 2026 notices kept, per the owner's rule — nothing is trademarked or registered yet, so ™/®/(TM)/(R) must not appear anywhere. Standing rule recorded for all Ametecs projects; old generated documents in docs\ still contain ™ inside binaries and will be regenerated clean if ever reissued.

7. Infrastructure Repairs (dev PC)

7.1 PHP 8.5.9 had no php.ini. Laragon's newly-added PHP 8.5.9 folder shipped without a php.ini, so PHP CLI loaded zero extensions and `migrate.bat` failed with "could not find driver". Created `php.ini` from the bundled development template with `extension_dir="ext"` and enabled: pdo_mysql, mysqli, pdo_sqlite + sqlite3 (needed by test.bat), curl, mbstring, openssl, fileinfo, intl, gd, zip, sodium. Both pending migrations then ran clean. The "Deprecated: PDO::MYSQL_ATTR_SSL_CA" lines during artisan runs are harmless PHP-8.5-vs-Laravel-11 noise, not errors (silenceable via error_reporting on request).

7.2 Git-over-mount procedure. Working through the device bridge, every git command leaves a stale `.git/index.lock` (the mount forbids unlink). Procedure recorded: move stale locks to `_to_delete\` before each git command, one command at a time. All lock files from this session are parked in `_to_delete\` for the owner to purge.

8. Standing Rules Recorded This Session

- 1. No ™/® anywhere** on Ametecs product names (nothing registered); "© 2026. All rights reserved." is allowed. All projects, all documents, all UI.
- 2. Never change composer/npm dependencies or versions without asking:** report current version → proposed → why, and wait for approval. (Nothing in this session touched composer.json.)
- 3. Signing-key override:** SmartEPT Central's licence keys are in git by owner decision — the repo must stay private; do not re-ignore the keys. Other products keep the never-in-git rule.
- 4. Seeder/password casts:** never `Hash::make()` into a 'hashed'-cast attribute; never set passwords via query-builder `update()`.

9. Deployment Checklist (owner)

Done during the session: Central migration (nullable price) ✓ · Product migration (mail_settings + password_otps) ✓ · php.ini fix ✓ · Razorpay verification of smartept.com ✓ · Central Settings: product URL + secrets filled ✓.

Local PC: Laragon Stop All → Start All; hard-refresh /admin, /buy and the landing page; product tinker unlock for `super@smartept.io` then change the password; set the global SMTP in product Ops → Email/SMTP and Send test; try Forgot password end-to-end; run Central `test.bat` (includes PerpetualProgressivePricingTest); `php artisan landing:import` in Central (then optional `landing:verify`); `git push` both repos.

Live VPS: `git pull` both apps; Central: `php artisan migrate --force` (tenants.status VARCHAR) + key permissions (`chmod 600` private key, `chown www-data`); Product: `php artisan migrate --force` + `config:clear` + reload php8.3-fpm + tinker unlock + set global SMTP; product .env: `PROVISION_SECRET` / `SSO_SHARED_SECRET` matching Central Settings; Interakt key into Settings → WhatsApp API (status active); confirm `APP_DEBUG=false`; re-run System Health — expect all green; watch the next top-of-the-hour dunning run comes through clean.

10. Open Items & Recommendations

- **Change every seeded live password now** (super + the four ametecs.io accounts) — highest-priority security action.
- Post-launch: revisit invoice Delete (restrict or switch to credit-note-only) per GST series compliance.
- Optional adds offered, not built: Super editing any company's SMTP via picker; portal coupon-box treatment; resend-OTP cooldown timer on trial signup; silencing PHP 8.5 deprecation notices.
- Backlog unchanged from the 7-Aug audit: secrets-at-rest encryption (S1), admin 2FA (S2), AMC-expiry dunning (B2), failed-payment webhooks, GST e-invoice IRN/QR.
- The companion **Test Case Suite** (Excel + PDF, same folder) contains 77 team-executable cases covering every change in this report.